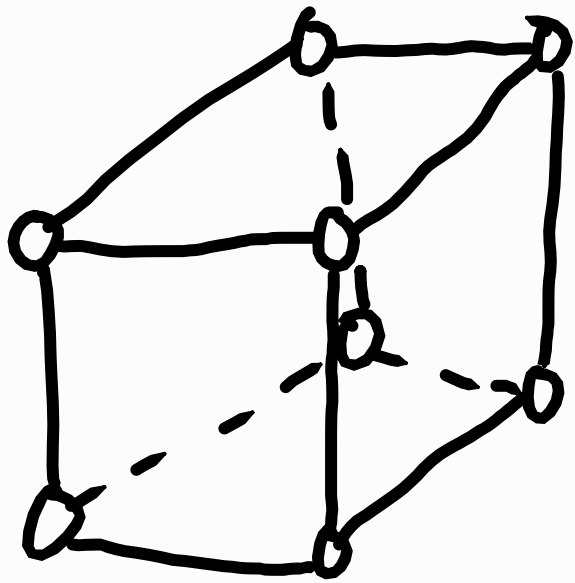
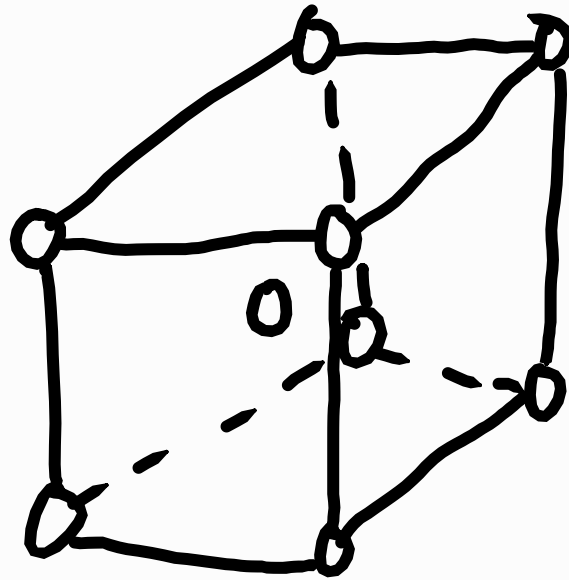


1) Simple Cubic



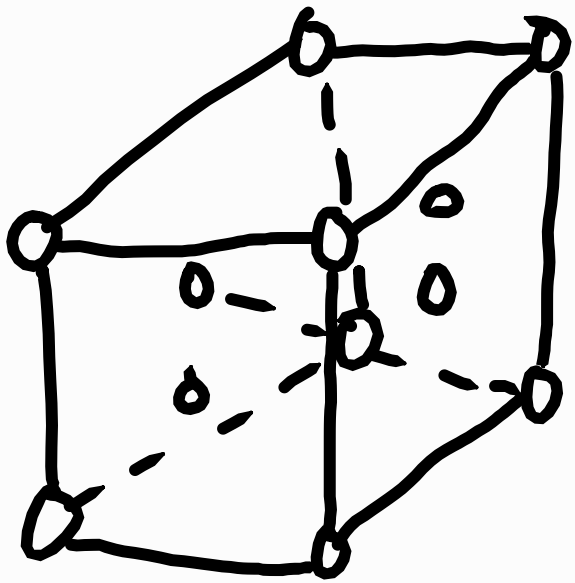
Coordination number: 6

Body-centred Cubic



Coordination number: 8

Face-centred Cubic



Coordination number: 12

2) Number of corner atoms: 8

Each corner atom is shared by 8 unit cells.

Number of edge atoms: 12

Each edge atom is shared by 4 unit cells.

$$\begin{aligned}\text{Number of atoms} &= 8 \times \frac{1}{8} + 12 \times \frac{1}{4} \\ &= 1 + 3 \\ &= 4\end{aligned}$$

3) Atomic packing fraction (APF)

$$= \frac{\text{Volume of atoms in a unit cell}}{\text{Total unit cell volume}}$$

For BCC:

$$a = \frac{4r}{\sqrt{3}}$$

$$\begin{aligned} \text{Volume of atoms} &= \left(1 + 8 \times \frac{1}{8}\right) \times \frac{4}{3} \pi r^3 \\ &= \frac{8}{3} \pi r^3 \end{aligned}$$

$$\begin{aligned} \text{Total unit cell volume} &= a^3 = \left(\frac{4r}{\sqrt{3}}\right)^3 \\ &= \frac{64r^3}{3\sqrt{3}} \end{aligned}$$

$$\begin{aligned} \text{Atomic packing fraction} &= \frac{\frac{8}{3} \pi \cancel{r^3}}{\frac{64 \cancel{r^3}}{3\sqrt{3}}} \\ &= \frac{\sqrt{3} \pi}{8} \\ &\approx 0.68 \end{aligned}$$

4) Density:

$$\rho_c = \frac{n \times A_m}{N_a \times V_c}$$

$$n = 1 + 8 \times \frac{1}{8}$$
$$= 2$$

$$A_m = 55.85 \text{ g mol}^{-1}$$

$$N_a = 6.023 \times 10^{23} \text{ atoms mol}^{-1}$$

$$a = \frac{4r}{\sqrt{3}}$$

$$r = 0.124 \text{ nm}$$

$$V_c = a^3 = \left(\frac{4(0.124 \times 10^{-9})}{\sqrt{3}} \right)^3$$

$$\approx 2.35 \times 10^{-29} \text{ m}^3$$

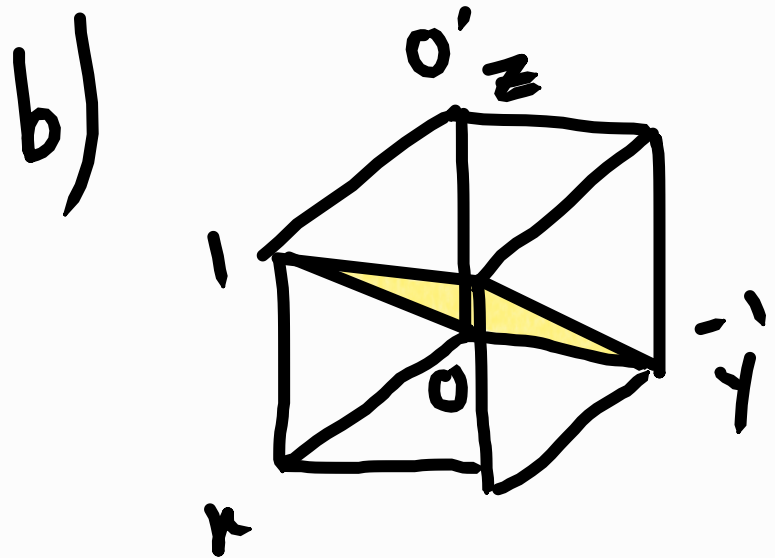
$$\rho_c = \frac{2 \times 55.85}{2.35 \times 10^{-29} \times 6.023 \times 10^{23}}$$

$$\approx 7897 \text{ kg m}^{-3}$$

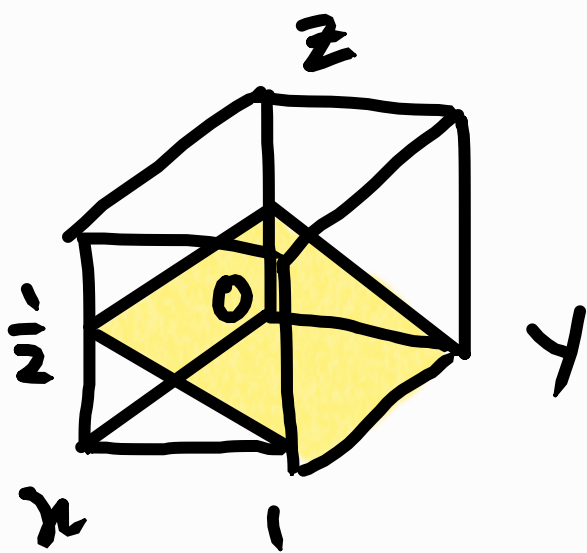
$$\approx 7.900 \text{ g cm}^{-3}$$

5a) Miller indices for A: $(\bar{1} \bar{1} 1)$

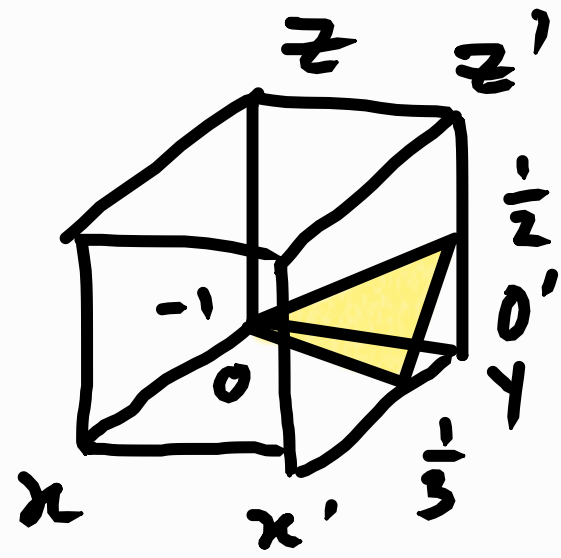
Miller indices for B: $(2 \ 3 \ 0)$



$(10\bar{1})$



(012)



$(3\bar{1}2)$

6a) A: $[1 \ 2 \ 0]$

B: $[1 \ 3 \ \bar{6}]$

C: $[\bar{4} \ 3 \ 0]$

$\frac{1}{2} \ 1 \ 0$

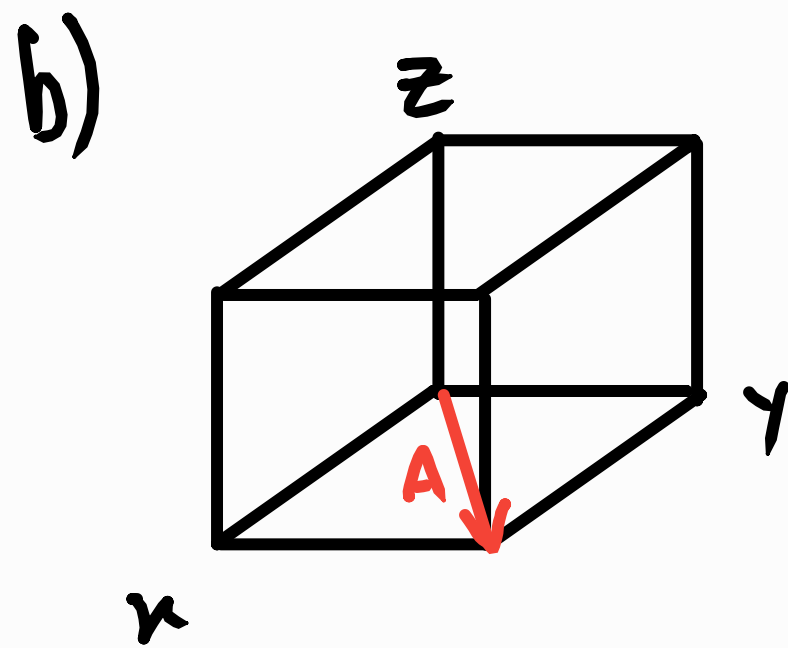
$\frac{1}{6} \ \frac{1}{2} \ -1$

$-\frac{2}{3} \ \frac{1}{2} \ 0$

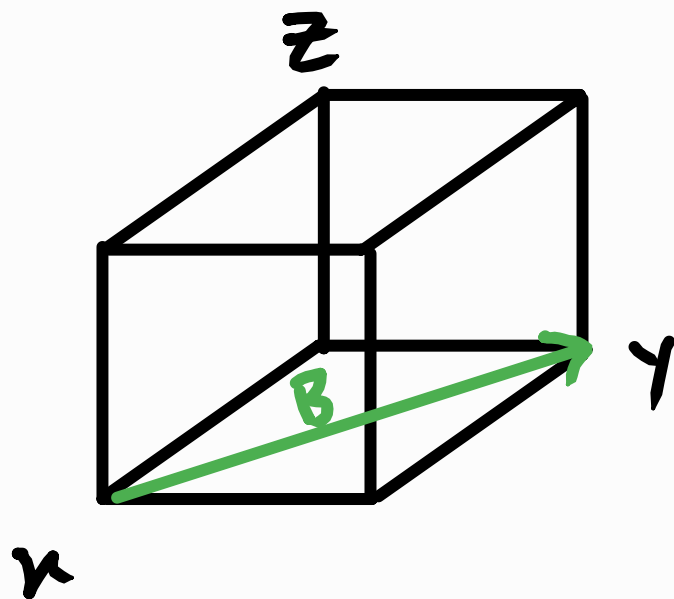
$1 \ 2 \ 0$

$1 \ 3 \ -6$

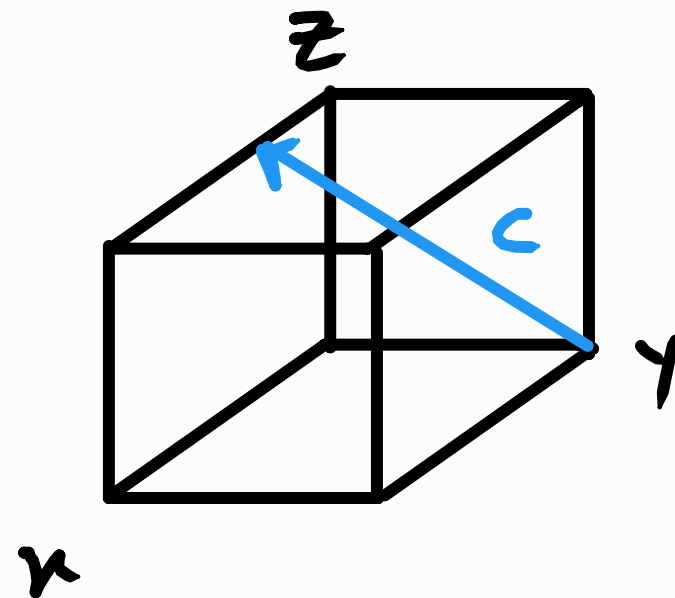
$-4 \ 3 \ 0$



$[110]$



$[\bar{1}10]$



$[1\bar{3}3]$
 $\frac{1}{3} \ -1 \ 1$

$$7) \text{ Linear density} = \frac{\text{No. of atoms centred on the vector}}{\text{Length of direction vector}}$$

For FCC $[111]$:

$$\begin{aligned} \text{Length of the vector} &= \sqrt{(4R)^2 + (2R\sqrt{2})^2} \\ &= \sqrt{24R^2} \\ &= 2\sqrt{6}R \end{aligned}$$

$$\text{No. of atoms} = \frac{1}{2} \times 2 = 1$$

$$\therefore LD_{[111]} = \frac{1}{2\sqrt{6}R}$$

$$\text{Silver } LD_{[111]} = \frac{1}{2\sqrt{6}(0.144)}$$

$$= 1.417528786$$

$$\approx 1.42 \text{ nm}^{-1}$$

$$7) \text{ Linear density} = \frac{\text{No. of atoms centred on the vector}}{\text{Length of direction vector}}$$

For FCC $[100]$:

$$\text{Length of the vector} = 2\sqrt{2}R$$

$$\text{No. of atoms} = \frac{1}{2} \times 2 = 1$$

$$LD_{[100]} = \frac{1}{2\sqrt{2}R}$$

$$\begin{aligned} \text{Silver } LD_{[100]} &= \frac{1}{2\sqrt{2}(0.144)} \\ &= 2.455231879 \\ &\approx 2.46 \text{ nm}^{-1} \end{aligned}$$

$$8) \text{ Planar density} = \frac{\text{Number of atoms centred on a plane}}{\text{Area of the plane}}$$

For FCC (100) :

$$\begin{aligned}\text{Area of the plane} &= a^2 \\ &= (2\sqrt{2}R)^2 \\ &= 8R^2\end{aligned}$$

$$\begin{aligned}\text{Number of atoms} &= \frac{1}{4} \times 4 + 1 \\ &= 2\end{aligned}$$

$$PD_{(100)} = \frac{2}{8R^2} = \frac{1}{4R^2}$$

$$\begin{aligned}\text{Ni: } PD_{(100)} &= \frac{1}{4(0.125)^2} \\ &= 16 \text{ nm}^{-2}\end{aligned}$$

$$8) \text{ Planar density} = \frac{\text{Number of atoms centred on a plane}}{\text{Area of the plane}}$$

For FCC (111) :

$$\begin{aligned} \text{Area of the plane} &= \frac{(4R)^2 \sqrt{3}}{4} \\ &= 4\sqrt{3}R^2 \end{aligned}$$

$$\begin{aligned} \text{Number of atoms} &= \frac{1}{6} \times 3 + 3 \times \frac{1}{2} \\ &= 2 \end{aligned}$$

$$PD_{(111)} = \frac{2}{4\sqrt{3}R^2} = \frac{2}{4\sqrt{3}R^2}$$

$$\begin{aligned} \text{Ni } PD_{(111)} &= \frac{2}{4\sqrt{3}(0.125)^2} \\ &= \frac{32\sqrt{3}}{3} \\ &\approx 18.48 \text{ nm}^{-2} \end{aligned}$$